Tested out sections -

Chapter 2, Mathematics of Machine Learning Book

2.1 $G = (\mathbb{R}\setminus\{-1\}, *)$

$a, b \in \mathbb{R}\setminus\{-1\}: a * b = ab + a + b$

(a) $a, b \in \mathbb{R}\setminus\{-1\} \Rightarrow a * b \in \mathbb{R}\setminus\{-1\}$

$\Rightarrow$ G is closed under * ($a*b = -1$ only when $a = b = -1$)

$$a*(b*c) = a*(bc+b+c)$$

$$= a(bc+b+c) + a + (bc+b+c)$$

$$= abc + ab + ac + bc + a + b + c$$

$$= (ab+a+b)c + ab + a + b + c$$

$$= (ab+a+b) * c = (a*b)*c$$

$\Rightarrow$ * satisfies associativity

$a*0 = 0+a+0 = a = 0*a$ and $0 \in \mathbb{R}\setminus\{-1\}$

$\Rightarrow$ 0 is the neutral element for *

$a*b = 0 \Rightarrow ab + a + b = 0$

$$\Rightarrow b(1+a) = -a \Rightarrow b = \frac{-a}{1+a}$$

$\Rightarrow \forall a \in \mathbb{R}\setminus\{-1\}, \exists\, b = \frac{-a}{1+a} \in \mathbb{R}\setminus\{-1\}$

such that $a*b = b*a = 0$ (inverse exists)

$a*b = ab+a+b = ba+b+a = b*a$

$\Rightarrow$ $*$ is commutative on $\mathbb{R}\setminus\{-1\}$

$\Rightarrow$ G is an Abelian Group.

(b) $3*x*x = 15$

$\Rightarrow 3*(x*x) = 15 \Rightarrow 3*(x^2+2x) = 15$

$\Rightarrow 3x^2+6x+(x^2+2x)+3 = 15$

$\Rightarrow 4x^2+8x = 12$ or $x^2+2x = 3 \Rightarrow x = 1$

2.3 $G = \left\{ \begin{bmatrix} 1 & x & z \\ 0 & 1 & y \\ 0 & 0 & 1 \end{bmatrix} \in \mathbb{R}^{3\times 3} \mid x,y,z \in \mathbb{R} \right\}$

- is matrix multiplication on $\mathbb{R}^{3\times 3}$

Let $x,y,z,a,b,c \in \mathbb{R}$ and

$A = \begin{bmatrix} 1 & x & z \\ 0 & 1 & y \\ 0 & 0 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{bmatrix} \in G$

$A \cdot B = \begin{bmatrix} 1 & a+x & b+cx+z \\ 0 & 1 & c+y \\ 0 & 0 & 1 \end{bmatrix} = B \cdot A \in G$

$\Rightarrow$ G is closed under $\cdot$

$\Rightarrow$ $\cdot$ is commutative on G

Let $C = \begin{bmatrix} 1 & p & q \\ 0 & 1 & r \\ 0 & 0 & 1 \end{bmatrix} \in G$

$$(A \cdot B) \cdot C = \begin{bmatrix} 1 & a+x & b+cx+z \\ 0 & 1 & c+y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & p & q \\ 0 & 1 & r \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & p+a+x & b+cx+z+q+ar+xr \\ 0 & 1 & c+y+r \\ 0 & 0 & 1 \end{bmatrix}$$

$$A \cdot (B \cdot C) = \begin{bmatrix} 1 & x & z \\ 0 & 1 & y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & a+p & b+ar+q \\ 0 & 1 & c+r \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & x+a+p & b+ar+q+z+cx+rx \\ 0 & 1 & c+r+y \\ 0 & 0 & 1 \end{bmatrix}$$

$\Rightarrow A \cdot (B \cdot C) = (A \cdot B) \cdot C$: $*$ is associative on G

$I \in \mathbb{R}^{3\times 3}$ and $I \in G$ is the identity element because $A \cdot I = I \cdot A = A \quad \forall A \in G$

Let $A \cdot B = I$

$$\Rightarrow \begin{bmatrix} 1 & a+x & b+cx+z \\ 0 & 1 & c+y \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$\Rightarrow a = -x,\ c = -y,\ b = xy - z$

$$\Rightarrow B = \begin{bmatrix} 1 & -x & xy-z \\ 0 & 1 & -y \\ 0 & 0 & 1 \end{bmatrix} \in G$$

$$\forall\ A = \begin{bmatrix} 1 & x & z \\ 0 & 1 & y \\ 0 & 0 & 1 \end{bmatrix} \in G$$

such that $A \cdot B = B \cdot A = I \Rightarrow$ inverse exists in G

$\Rightarrow G$ is Abelian group

2.13 $A_1 = \begin{bmatrix} 1 & 0 & 1 \\ 1 & -2 & -1 \\ 2 & 1 & 3 \\ 1 & 0 & 1 \end{bmatrix}$ $A_2 = \begin{bmatrix} 3 & -3 & 0 \\ 1 & 2 & 3 \\ 7 & -5 & 2 \\ 3 & -1 & 2 \end{bmatrix}$

$U_1 : \{x \mid A_1 x = 0\}$ and $U_2 : \{x \mid A_2 x = 0\}$

$$\text{RREF}(A_1) = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\Rightarrow U_1 = \left\{ \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} \right\} \Rightarrow \dim(U_1) = 1$$

$$\text{RREF}(A_2) = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\Rightarrow U_2 = \left\{ \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} \right\} \Rightarrow \dim(U_2) = 1$$

(a) $\dim(U_1) = \dim(U_2) = 1$

(b) $U_1 = \left\{ \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} \right\}$, $U_2 = \left\{ \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} \right\}$

(c) $U_1 \cap U_2 = \left\{ \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} \right\}$

2.18 f and g are automorphisms on E, such that

$f \circ g = id_E$

Since f and g are automorphisms:

$\ker(f) = \ker(g) = \{0\} \Rightarrow \ker(g \circ f) = \{0\}$

$\operatorname{Im}(g) = \operatorname{Im}(f) = E \Rightarrow \operatorname{Im}(g \circ f) = \operatorname{Im}(g)$

and $\ker(f) \cap \operatorname{Im}(g) = \{0\}$

2.20 $B = \begin{bmatrix} 2 & -1 \\ 1 & -1 \end{bmatrix}$, $B' = \begin{bmatrix} 2 & 1 \\ -2 & 1 \end{bmatrix}$

(b) $P_1 : B' \to B$ is given by $B^{-1}B' = \begin{bmatrix} 4 & 0 \\ 6 & -1 \end{bmatrix}$

(c) (i) $C = \begin{bmatrix} 1 & 0 & 1 \\ 2 & -1 & 0 \\ -1 & 2 & -1 \end{bmatrix}$

$\det(C) = 1.1 + 1(4-1) = 4 \neq 0 \Rightarrow C$ is a basis of R^3

(ii) $C' = I_{3\times 3}$

$P_2 : C \to C'$ is given by $C'^{-1}C = C$

(d) $\phi(b_1+b_2) = c_2+c_3$

$\phi(b_1-b_2) = 2c_1-c_2+3c_3$

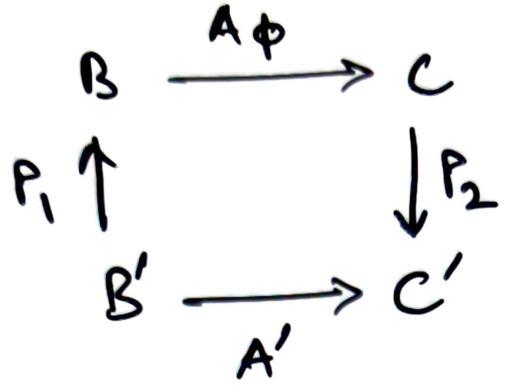
$\Rightarrow \phi(b_1) = \frac{1}{2}(c_2+c_3+2c_1-c_2+3c_3) = c_1+2c_3$

$\Rightarrow \phi(b_2) = \frac{1}{2}(c_2+c_3-2c_1+c_2-3c_3) = -c_1+c_2-c_3$

$\Rightarrow A\phi = [\phi(b_1) \ \phi(b_2)]$

$= \begin{bmatrix} 3 & -2 \\ 2 & -3 \end{bmatrix}$

(e)



$$\Rightarrow A' = P_2 A\phi P_1 = C A\phi P_1$$

$$= \begin{bmatrix} 1 & 0 & 1 \\ 2 & -1 & 0 \\ -1 & 2 & -1 \end{bmatrix} \begin{bmatrix} 3 & -2 \\ 2 & -3 \\ -3 & 4 \end{bmatrix} \begin{bmatrix} 4 & 0 \\ 6 & -1 \end{bmatrix} = \begin{bmatrix} 12 & -2 \\ 10 & 1 \\ -32 & 8 \end{bmatrix}$$

(f) $x_{B'} = 2b_1' + 3b_2' = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$

(i) $x_B = P_1 x_{B'} = \begin{bmatrix} 4 & 0 \\ 6 & -1 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 8 \\ 9 \end{bmatrix}$

(ii) $\phi(x)_C = A_\phi x_B = \begin{bmatrix} 3 & -2 \\ 2 & -3 \\ -3 & 4 \end{bmatrix} \begin{bmatrix} 8 \\ 9 \end{bmatrix} = \begin{bmatrix} 6 \\ -11 \\ 12 \end{bmatrix}$

(iii) $\phi(x)_{C'} = C\,\phi(x)_C = \begin{bmatrix} 18 \\ 23 \\ -40 \end{bmatrix}$

(iv) $\phi(x)_{C'} = A' x_{B'} = \begin{bmatrix} 12 & -2 \\ 10 & 1 \\ -32 & 8 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 18 \\ 23 \\ -40 \end{bmatrix}$

3.6 $\langle x, y \rangle = x^T A y$ with $A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$

(a) $V = \text{span}[e_1, e_3]$. Let $v = \pi_V(e_2) \in V$

Let $B = [e_1\ e_3]$ be a basis of V then $v = B\lambda$

$\langle v, e_2 - v \rangle = 0 \Rightarrow v^T A e_2 = v^T A v$

$\Rightarrow v^T A (e_2 - v) = 0 \Rightarrow \lambda^T B^T A (e_2 - v) = 0$

since $e_2 - v \neq 0 \Rightarrow e_2 - v = p \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}$ where

Nullspace of $B^T A = \left\{ \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix} \right\}$

$\Rightarrow v = e_2 - p \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix} = e_2 - p\alpha$ where

$\alpha^T = [-1\ 2\ 1]$

$\Rightarrow (e_2 - p\alpha)^T A p\alpha = 0$

$\Rightarrow e_2^T A \alpha = p \alpha^T A \alpha \Rightarrow p = \dfrac{e_2^T A \alpha}{\alpha^T A \alpha} = \dfrac{1}{2}$

$\Rightarrow v = e_2 - \dfrac{1}{2}\alpha = \begin{bmatrix} 1/2 \\ 0 \\ -1/2 \end{bmatrix}$

(b) $d(e_2, U) = \sqrt{\langle e_2 - v, e_2 - v \rangle}$

$= \sqrt{\langle \frac{1}{2}\alpha, \frac{1}{2}\alpha \rangle} = \frac{1}{2}\sqrt{\alpha^T A \alpha} = 1$

3.7 (a) $id_V - \pi$ is a projection

$\Leftrightarrow (id_V - \pi) \circ (id_V - \pi)(x) = (id_V - \pi)(x)$

$\Leftrightarrow (id_V \circ id_V - 2\pi \circ id_V + \pi \circ \pi)(x)$

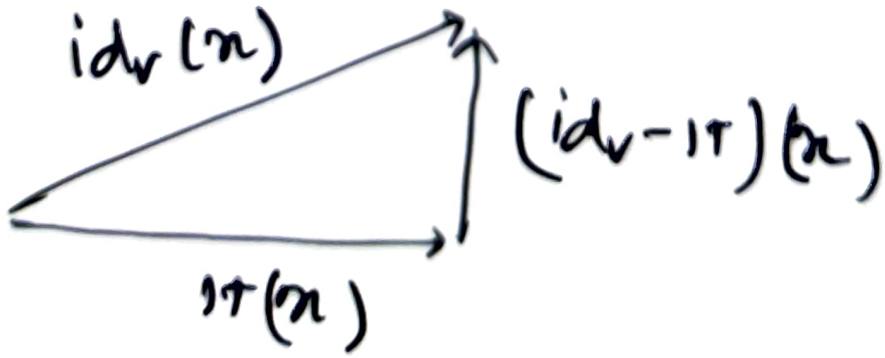
$= (id_V - \pi)(x)$

$\Leftrightarrow (id_V - 2\pi + \pi \circ \pi)(x) = (id_V - \pi)(x)$

$\Leftrightarrow (\pi \circ \pi)(x) = \pi(x)$

$\Leftrightarrow \pi$ is a projection

(b)



$\ker(\pi) = \mathrm{Im}(id_V - \pi)$

$\mathrm{Im}(\pi) = \ker(id_V - \pi)$

3.9 $x_1, x_2, \cdots, x_n > 0$ with

$$x_1 + x_2 + \cdots + x_n = 1$$

Let $x = [x_1, x_2, \cdots, x_n]^T$ and

$$\mathbb{1} = [1, 1, \cdots, 1]^T$$

$$\Rightarrow \mathbb{1}^T x = x_1 + x_2 + \cdots + x_n = 1$$

$$\leq \|\mathbb{1}\| \cdot \|x\| = \sqrt{n} \cdot \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}$$

$$\Rightarrow x_1^2 + x_2^2 + \cdots + x_n^2 \geq \frac{1}{n} \quad \text{—— (a)}$$

Let $a = [\sqrt{x_1}, \sqrt{x_2}, \cdots, \sqrt{x_n}]^T$ and

$$b = \left[1/\sqrt{x_1}, 1/\sqrt{x_2}, \cdots, 1/\sqrt{x_n}\right]^T$$

$$\Rightarrow a^T b = n \leq \|a\| \cdot \|b\| = 1 \cdot \sqrt{\frac{1}{x_1} + \frac{1}{x_2} + \cdots + \frac{1}{x_n}}$$

$$\Rightarrow \frac{1}{x_1} + \frac{1}{x_2} + \cdots + \frac{1}{x_n} \geq n^2 \quad \text{—— (b)}$$